

Advanced Lesson 11 - Exercises:

Patterns in Arguments and in Replacement Rules

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MATH 7013 - Advanced Mathematical Modeling

1: Exactly one argument, squared

```
ClearAll[f, x, y];  
f[x_] := x2  
?  
f[2], f[x], f[y], f["string"], f[{1, 2, 3}]  
f[{1, 2}, {3, 4}]  
{f[], f[x, y], f[1, 2, 3, 4]}
```

2: Exactly zero arguments

```
ClearAll[f, x, y];  
f[] := 3  
?  
f[]  
{f[2], f[x], f[x, y], f[1, 2, 3, 4], f[{1, 2}, {3, 4}]}
```

3: One or more arguments, squaring the list of them

```
ClearAll[f, x, y];
f[x__] := {x}2
?f
{f[2], f[x], f[y], f["string"], f[{1, 2, 3}]}
```

```
{f[{1, 2}, {3, 4}], f[x, y], f[1, 2, 3, 4]}
f[x, {1, 2, 3}]
f[x, {1, 2, 3}, {{1, 2}, {3, 4}}]
f[]
```

4: Zero or more arguments

```
ClearAll[f, x, y];
f[x___] := {x}2
?f
{f[2], f[x], f[y], f["string"], f[{1, 2, 3}]}
```

```
{f[{1, 2}, {3, 4}], f[x, y], f[1, 2, 3, 4]}
f[x, {1, 2, 3}]
f[]
```

5: Exactly two arguments

```
ClearAll[f, x, y];
f[x_, y_] := x2 Sin[y]
?f
{f[], f[x], f[{1, 2}], f[{1, 2, 3}], f[1, 2, 3, 4]}
```

```
f[x, y]
f[x, {1, 2, 3}]
f["string", {{1, 2}, {3, 4}}]
```

6: Exactly three, and exactly four, arguments

```
ClearAll[f3, f4, x, y, z, w];
f3[x_, y_, z_] := {x, y, z};
f4[x_, y_, z_, w_] := {x, y, z, w}

{f3[1, 2, 3], f4[1, 2, 3, 4]}

{f3[1, 2], f4[1, 2, 3]}
```

7: One argument, then zero or more

```
ClearAll[f, x, y, z, w, A, B];
f[x_, y___] := x2 Sin[{y}]

?f

f[]
f[x]
f[{1, 2}]
f[x, y]
f[2, 3, 4, 5]
f[x, y, z, w]
f[x, {1, 2, 3}, A, B]
ColumnForm[f[x, {1, 2, 3}, A, B]]
f["string", {{1, 2}, {3, 4}}]
```

8: Matching inside a Sine

```
ClearAll[f, x, y];
f[Sin[x_]] := x2

?f

{f[Sin[2]], f[Sin[y]], f[Sin[Log[x]]]}

f[Sin[{1, 2, 3}]]

{f[x], f[Cos[y]], f[1, 2, 3, 4]}
```

9: A derivative table by pattern matching

```
ClearAll[f, x, y];
f[Sin[x_]] := Cos[x];
f[Cos[x_]] := -Sin[x];
f[Log[x_]] :=  $\frac{1}{x}$ ;
f[e^x_] := e^x
?f

{f[e^x], f[Sin[y]], f[Cos[x + y]], f[Sin[2]]}

f[Sin[Log[x]]]

f[Sin[{1, 2, 3}]]
```

10: Odd functions, by a recursive pattern

```
ClearAll[f, x, y];
f[-x_] := -f[x]
?f

{f[-x], f[-y], f[-Sin[y]]}

f[-5]

{FullForm[-5], FullForm[-x]}
```

11: Extending the rule to negative numbers

```
ClearAll[f, x, y];
f[-x_] := -f[x];
f[n_?Negative] := -f[-n]
?f

{f[-x], f[x], f[-y], f[-Sin[y]]}

{f[-5], f[-7], f[5]}
```

12: The third element of a list of three or more

```
ClearAll[f, x, y, A, B];
f[{_, _, c_, ___}] := c
?f
```

```
{f[], f[x], f[{1, 2}]}
{f[{1, 2, 3}], f[{1, 2, 3, 4}], f[{1, 2, 3, 4, 5}]}
f[{x, y, {1, 2, 3}, A, B}]
```

13: Returning the first inner list from a list

```
ClearAll[f, x, y, A, B];
f[____, l_List, ____] := l
?f
f[1, 2, {4, 5}, 6]
f[{1, 2, 3, 4}]
f[{1, 2, {}, 3, 4}]
{f[{1, 2, 3, 4}], f[{1, 2, {3}, 4}], f[{1, {2, 3}, 4}], f[{1, 2, 3, {4}}]}
{f[{1, 2, {3, 4, 5}}], f[{1, 2, 3, 4, 5}]}
f[{1, 2, {3, {4, 5}}}]
f[{x, y, {1, 2, 3}, A, B}]
```

14: Full forms of patterns, and a positive-integer function

```
Clear[x];
Column[List @@ Function[u, ToString[FullForm[u]]] /@
  Hold[_, x_, x_, x_, x_Integer, x_?EvenQ, x_? (#1 > 3 &), x_Integer?EvenQ]]
ClearAll[f];
f[x_Integer?Positive] := "Positive Integer"
{f[3], f[3.45`], f[-3]}
```

15: An immediate rule replacing 2 with 222

```
ClearAll[rule1, x];
rule1 = 2 → 222
{1, 2, 3, 4} /. rule1
Sin[2] /. rule1
{1, 2, Sin[2], x^2} /. rule1
{1, 2, 3, 2 + 2} /. rule1
{1, 2, 3, Hold[2 + 2]} /. rule1
```

```
2 Sin[2 Sin[2 Sin[2 Sin[2 x]]]] /. rule1
```

16: An immediate rule on a cube

```
ClearAll[rule2, x, y, z];
rule2 = x_3 → x333

{x2, x3, x4, y3, Sin[x]3,  $\frac{1}{2} + z^3$ , (x + y)3} /. rule2

 $\left(\frac{1}{2} + y\right)^3$  /. rule2

{FullForm[x3], FullForm[(2 + y)3], FullForm[ $\left(\frac{1}{2} + y\right)^3$ ]}

Hold[ $\left(\frac{1}{2} + y\right)^3$ ] /. rule2
```

17: Two rules in a list, and why the first one gets missed

```
{x2, x3, x4, y3, Sin[x]3,  $\frac{1}{2} + z^3$ , (x + y)3} /. {rule1, rule2}

(2 + y)3 /. {rule1, rule2}

(2 + y)3 //. {rule1, rule2}
```

18: Turning a plotted curve into a cloud of points

```
ClearAll[x, plot1, plot2, plot3];
plot1 = Plot[x2, {x, 0, 1}]

Length[Cases[plot1, _Line, ∞]]

plot2 = plot1 /. Line[{pts__}] → Line[Point /@ {pts}];
Length[Cases[plot2, _Point, ∞]]

plot3 = plot2 /. Line[{pts__Point}] → {pts};
{Head[plot3], Length[Cases[plot3, _Line, ∞]], Length[Cases[plot3, _Point, ∞]]}

plot3
```